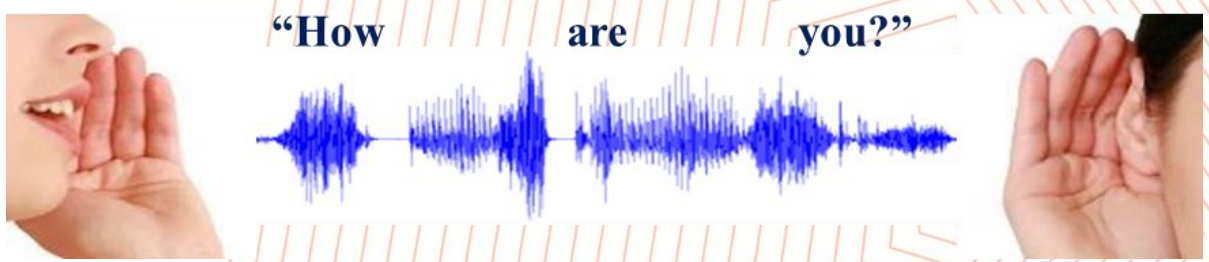
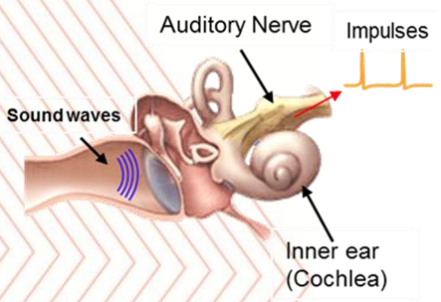
# ELEC3104: Mini-Project – Cochlear Signal Processing




Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

# ELEC3104: Project Outline

✓ This **individual** mini-project focuses on understanding and modelling the spectral analysis performed by the human cochlea.

**Introduction**

- Introduction to the Human Auditory System

**Mini-Project: Pass-Level**

- Implement a cascaded filter bank model of the cochlea for spectral analysis

**Mini-Project: Higher-Level**

- Incorporate nonlinear mechanisms into the cascaded cochlear model to estimate the pitch period of voiced speech

- **Additional Information:** You are encouraged to explore animations and videos explaining how the peripheral auditory system works, particularly the cochlea. These visual resources will help you understand cochlear signal processing.

  Eg: Cochlear Animation - https://www.youtube.com/watch?v=dyenMluFaUw

# ELEC3104: Mini-Project – Cochlear Signal Processing

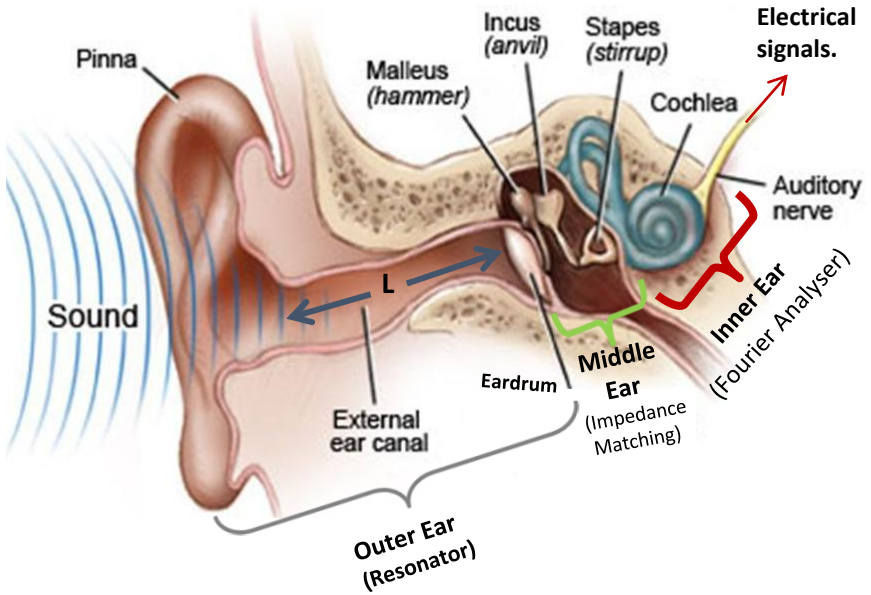
Introduction to the Human Auditory System

Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

# Introduction to the Human Auditory System

- ✓ The human auditory system is responsible for converting pressure variations caused by the sound waves that reach the ear into nerve impulses that are interpreted by the brain.
- ✓ The Human Auditory System is designed to assess frequency (pitch) and amplitude (loudness).
- ✓ The peripheral auditory system is divided into the Outer Ear, Middle Ear, and Inner Ear.
- ✓ The peripheral auditory system and in particular the cochlea can be viewed as a real-time spectrum analyser.
- ✓ The primary role of the cochlea is to transform the incoming complex sound wave at the ear drum into electrical signals.
- ✓ The human ear can respond to minute pressure variations in the air if they are in the audible frequency range, roughly 20 Hz - 20 kHz

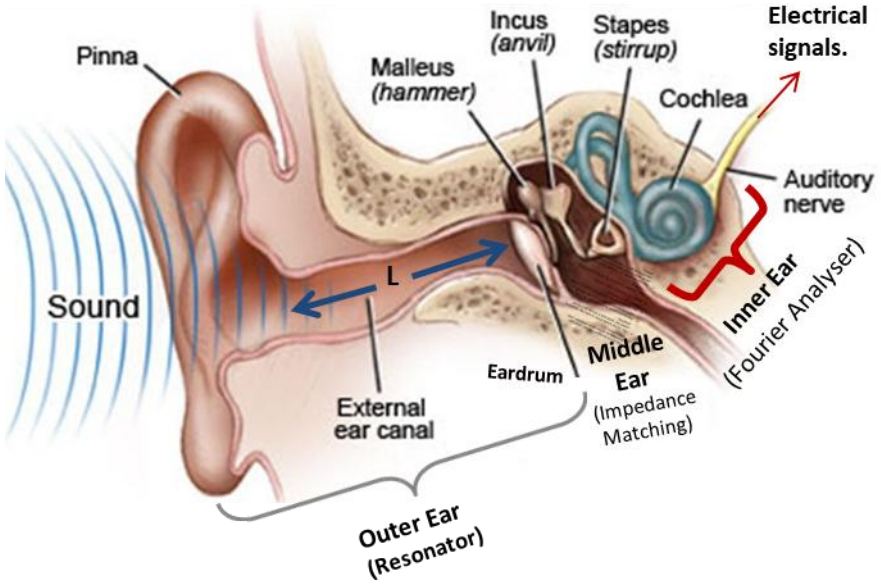


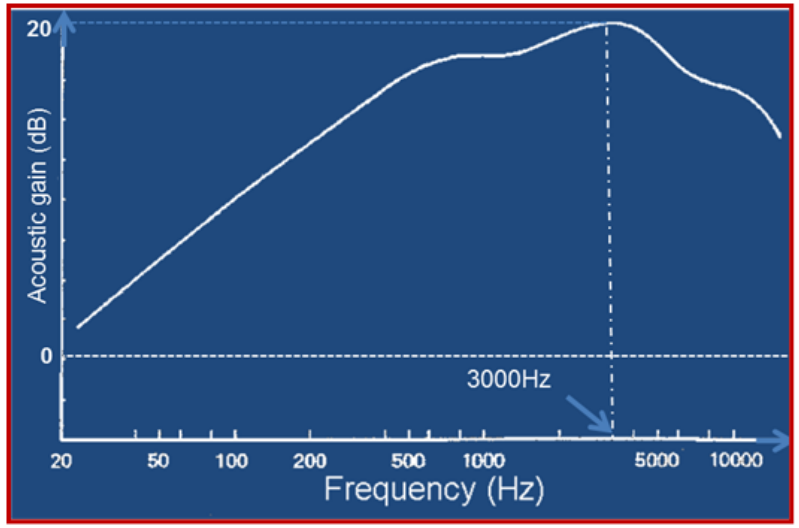

| Sounds | Level |
|---|---|
| Faint | 20dB (A faint Whisper is 30dB) |
| Soft (Quiet) | 40dB |
| Moderate | 60dB (normal conversation) |
| Loud | 80dB (alarm clocks, vacuum cleaners) |
| Very Loud | 90dB(Blenders);110dB (Concerts, car horns) |
| Uncomfortable | 120dB (jet planes during take off) |
| Painful and dangerous | 130dB(Jackhammers); 140dB(Gunshots)<br>*Use hearing protection |

- ✓ Over 85 dB for extended periods can cause permanent hearing loss
- ✓ Zero decibels (0 dB) represent the absolute threshold of human hearing, below which we cannot hear a sound.

# Outer Ear (Air Vibration): A resonator

- ✓ The pinna surround the ear canal and functions as sound wave reflectors and attenuators .
- ✓ The sound waves enter a tube-like structure called ear canal and it serves as a sound amplifier.
- ✓ The sound waves travel through the canal and reach the eardrum and cause it to vibrate
- ✓ The length (L) of the human ear canal is 2.8 cm (and 7 mm in diameter)
- ✓ Speed of sound (c) = 340.3 m/sec ;
- ✓ The resonant frequency (f) of the canal is = $\frac{c}{4L}$ = 3,038Hz.
- ✓ The human outer ear is most sensitive at about 3kHz and provides about 20dB (decibels) of gain to the eardrum at around 3000Hz.

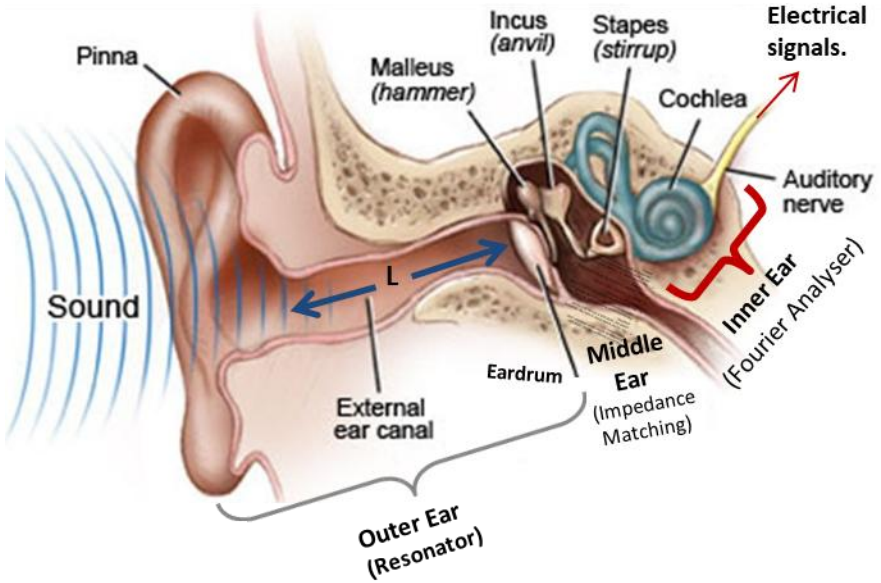


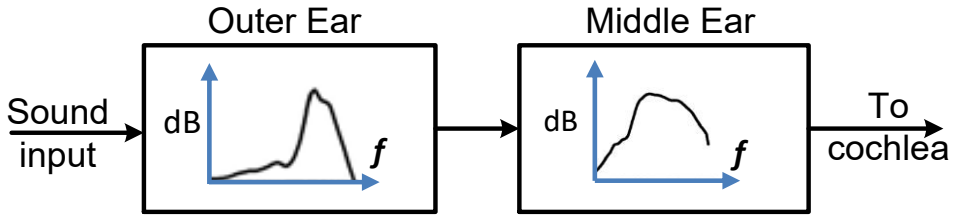




Outer ear is a low-Q bandpass filter
(Representative figure only)

# Middle Ear: An Impedance Matcher & an Amplifier

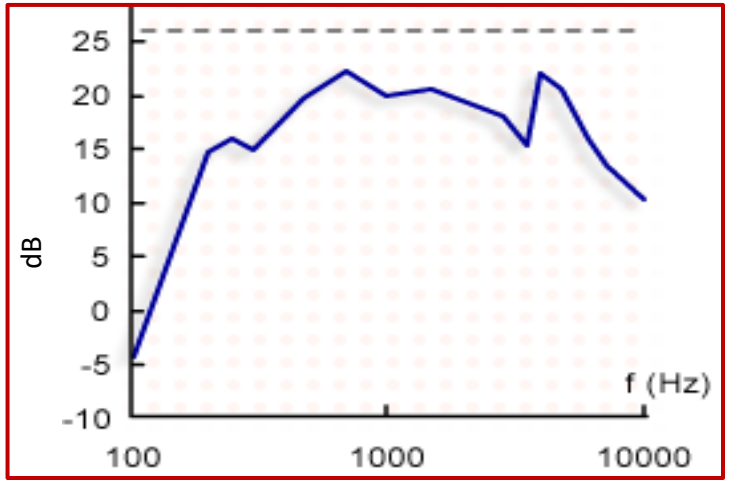
- ✓ Middle ear transforms the vibrating motion of the eardrum into motion of the stapes via the two tiny bones, the malleus and incus .
- ✓ The pressure of the sound waves on the oval window is around 25 times higher than on the eardrum.
- ✓ Since the sound Intensity ($I$) is proportional ($\propto$) to the square pressure ($P^2$) , the sound intensity increases 625 times (or 28dB)
- ✓ Middle ear converts acoustic energy to mechanical energy and mechanical energy to hydraulic energy



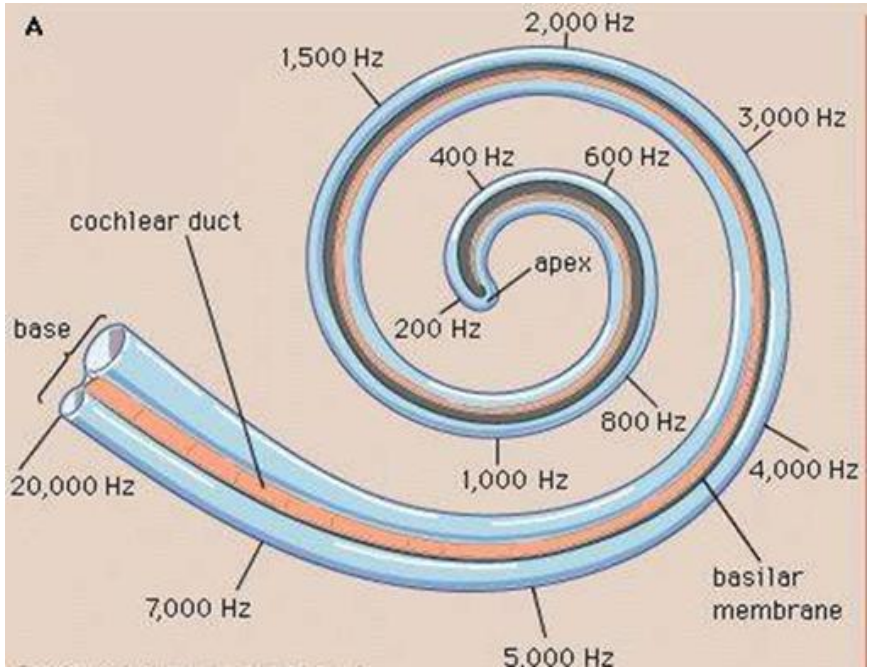

The combined frequency response of the outer and middle ear is a band-pass response, with its peak dominated near 3 kHz

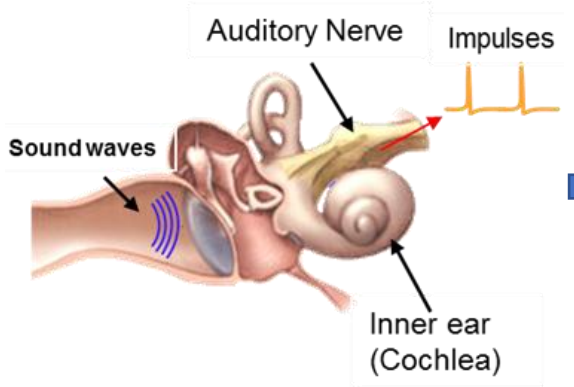
**Middle Ear Gain function**

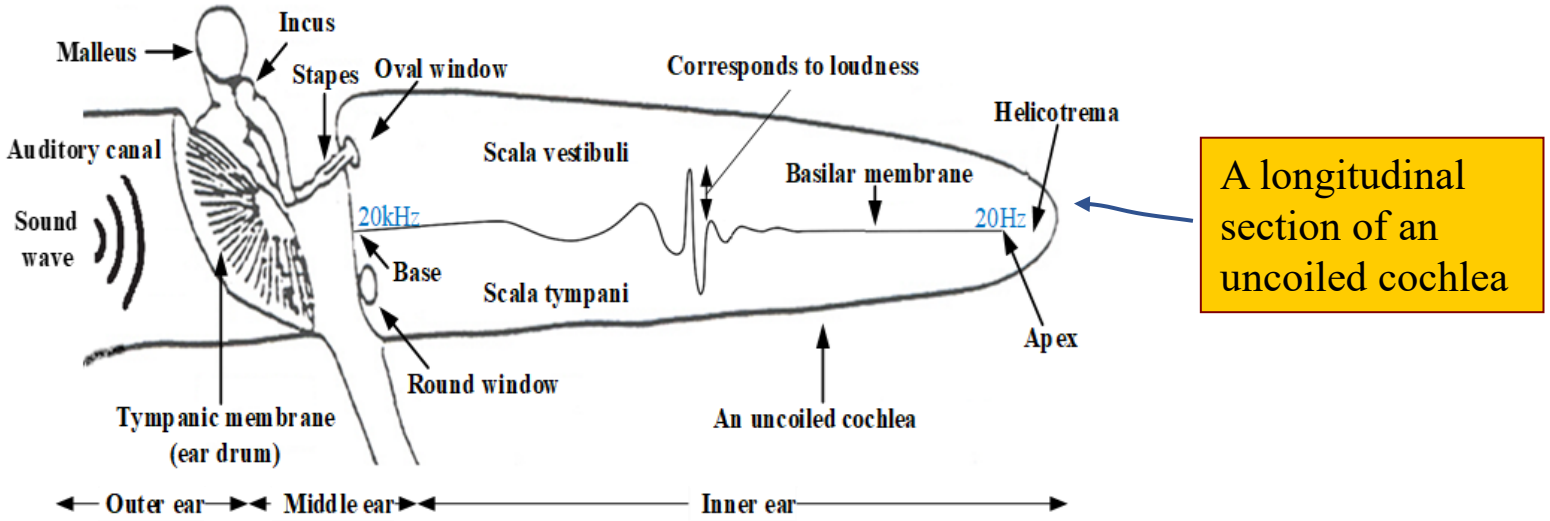


# Inner Ear

- ✓ The **inner ear** consists of the cochlea responsible for converting the vibrations of sound waves into electrochemical impulses which are passed on to the brain via the auditory nerve.
- ✓ The cochlea is a spiral shaped structure which is about 3.5 cm in length if uncoiled.
- ✓ The cochlea is divided along its length by the basilar membrane (BM) which partitions the cochlear into two fluid canals (scala vestibuli and scala tympani).
- ✓ The BM terminates just reaching the helicotrema, so there is a passage way between the scala vistibuli and the scala tymapni equalising the difference in pressure at the ends of the two scalas.



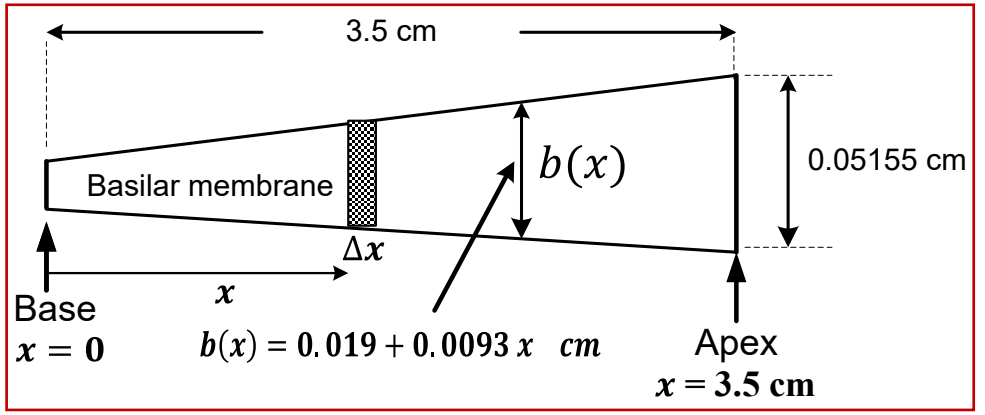




A longitudinal section of an uncoiled cochlea

# Basilar Membrane (Hydro Dynamical process)

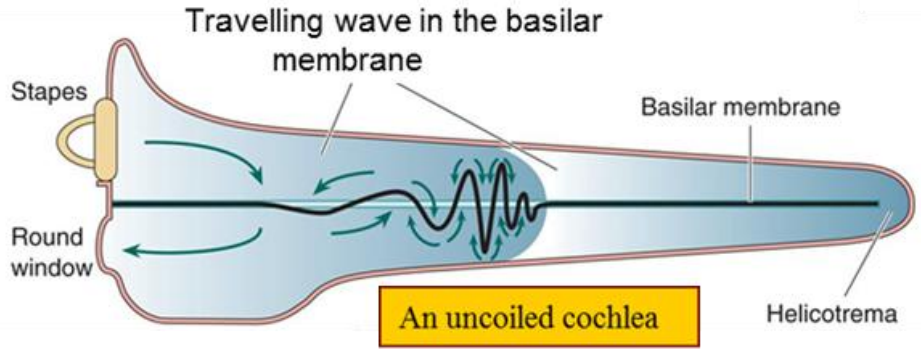
- ✓ The Basilar Membrane varies in width and stiffness along its length.
- ✓ At basal end it is narrow and stiff where as towards the apex it is wider and more flexible.
- ✓ Each point along the basilar membrane has a characteristic frequency, $f_p(x)$, to which it is most responsive.
- ✓ The maximum membrane displacement occurring at the basal end for high frequencies (20 kHz) and at the apical end for low frequencies (70Hz) .
- ✓ When the vibrations of the eardrum are transmitted by the middle ear into movement of the stapes, the resulting pressure differences between the cochlear fluid chambers, generate a travelling wave that propagates down the cochlea and reach maximum amplitude of displacement on the basilar membrane at a particular point before slowing down and decaying rapidly
- ✓ The location of the maximum amplitude of this travelling wave varies with the frequency of the eardrum vibrations




If $x$ is the distance of a point on the basilar membrane from the stapes, then the frequency, $f_p(x)$, that produces a peak at this point is given by:

$$f_p(x) = (20000.0)\,10^{-0.667\,x}\,Hz \quad 0 \le x \le 3.5\,cm$$

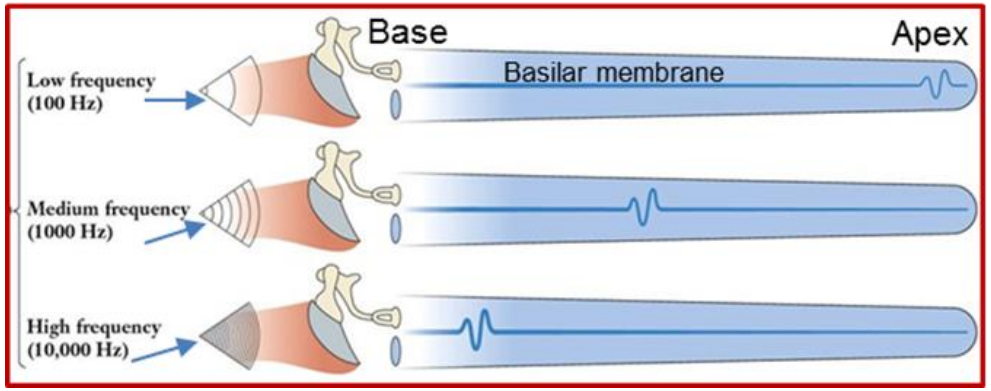
- It is evident that a 20 kHz tone at the stapes will cause the BM to vibrate at a point $x = 0$.
- A 70 Hz tone will excite the BM at a point x = 3.5 cm (i.e. at the apex)

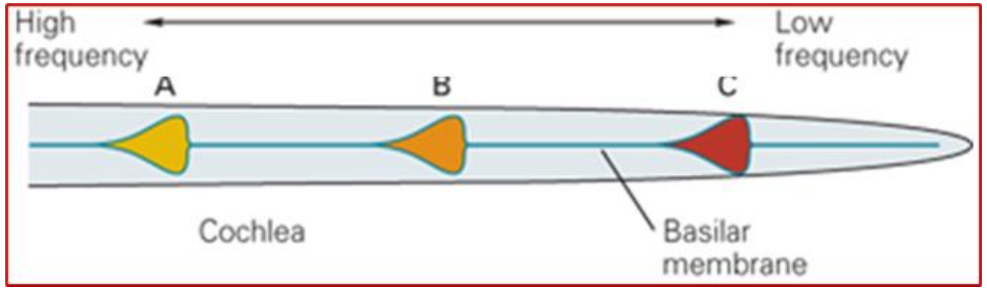


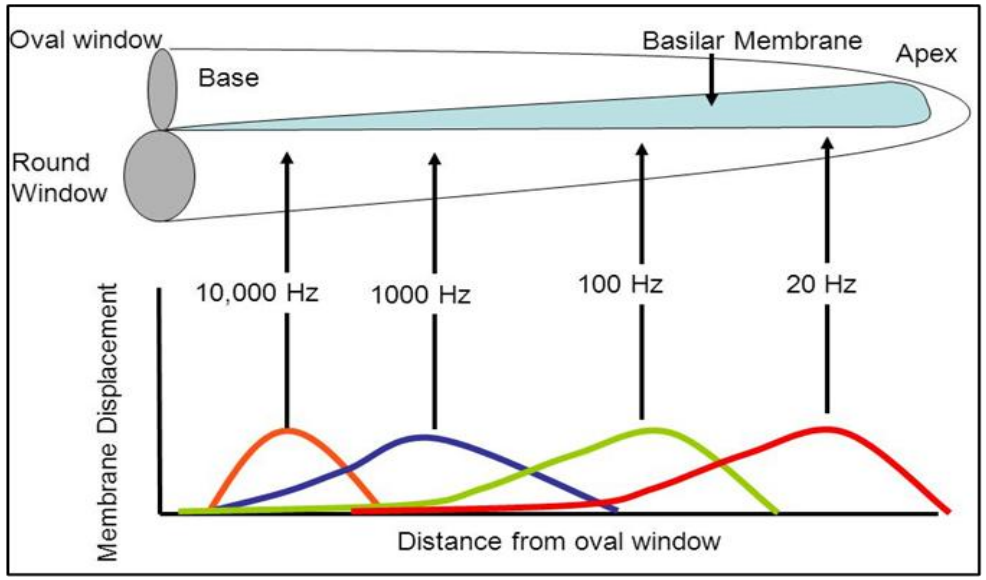

The basilar membrane is a resonant structure that vibrates, vertically in sympathy with pressure variations in the cochlear fluid.

# Basilar Membrane

- ✓ Different frequencies stimulate different areas of the basilar membrane
- ✓ When a tone (single sinusoid) is applied, the cochlear fluid oscillates in phase with the stimulating frequency causing a travelling wave pattern of the vibration on the basilar membrane
- ✓ There will be one place where the resonant frequency of the membrane matches the stimulus frequency and this place will show the maximum amount of vibration
- ✓ By measuring vibration at particular points on the membrane for a range of stimulus frequencies we can plot the frequency response of each place on the membrane
- ✓ The essential function of the basilar membrane is to act as a frequency analyser (a set of band-pass filters each responding to a different frequency region) resolving an input sound at the eardrum into its constituent frequencies

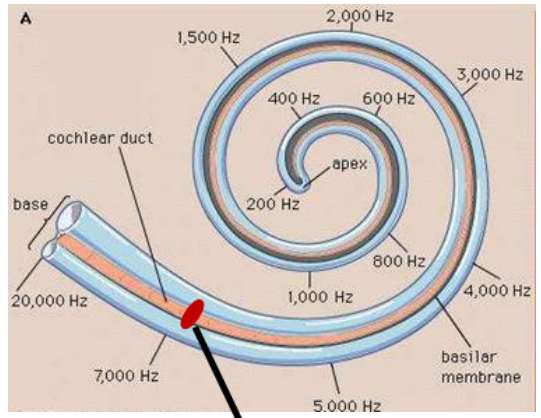


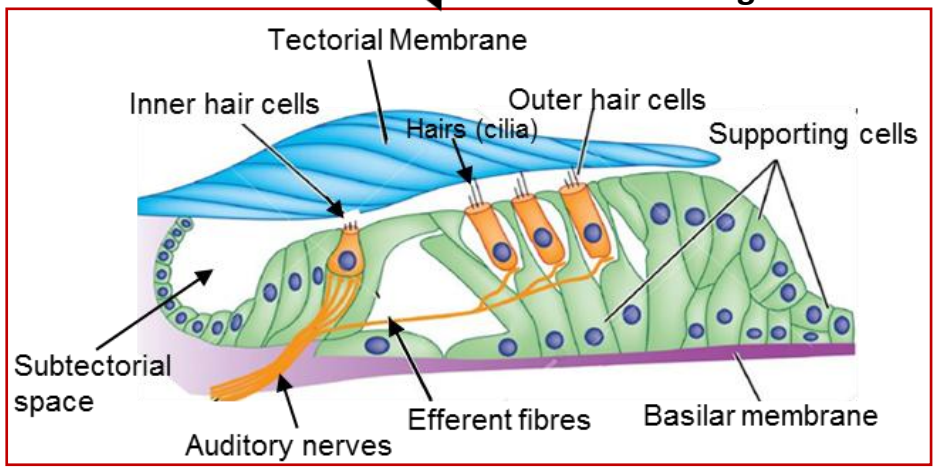



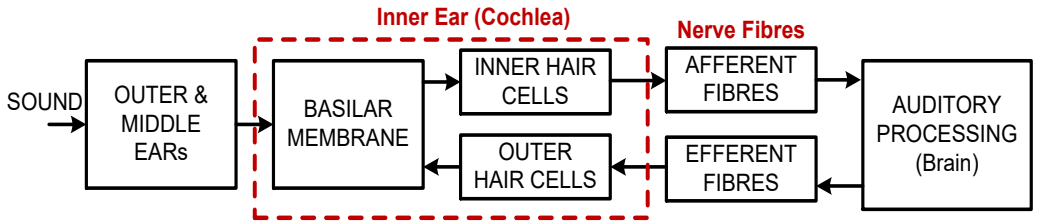
# Organ of Corti

- ✓ Attached to the basilar membrane and running its entire length is the organ of corti containing some 30,000 sensory hair cells.
- ✓ The hairs (cilia) of these cells stick up from the organ of corti and are in contact with overlying Tectorial Membrane
- ✓ There are two types of sensory hair cells:
  - One row of inner hair cells, whose cilia float freely in the fluid-filled region called subtectorial space
  - Three rows of outer hair cells whose cilia are attached to the tectorial membrane
- ✓ Most of the afferent fibres (neurons which carry signals to the brain) come from inner hair cells,
- ✓ The efferent fibres (which receive signals from the brain) go mainly to outer hair cells.
- ✓ When the basilar membrane deflects, due to pressure wave in the cochlear fluid, the tectorial membrane move and shear which causes the hairs of the outer hair cells to bend and also cause the fluid flow in the subtectorial space.
- ✓ This in turn triggers the inner hair cells to transmit nerve impulses along the afferent fibres and eventually to brain.
- ✓ The motion of each part of the basilar membrane as detected by the inner hair cells is transmitted as neural description to the brain.




**Organ of Corti**

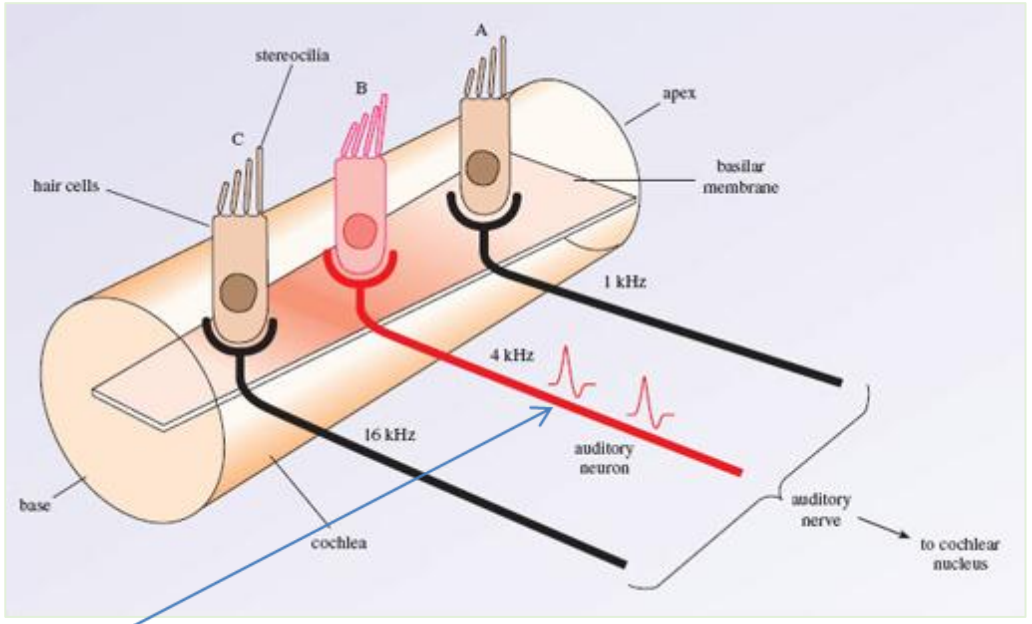


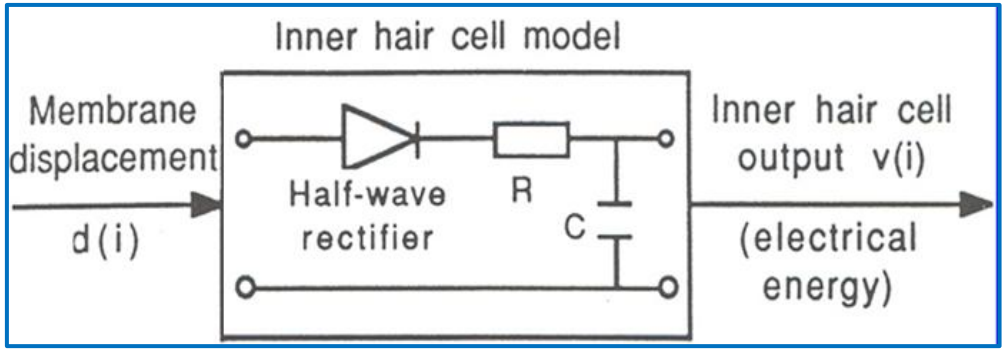

**A simplified Diagram of a Human Auditory System**

# Mechanical to Neural Transduction (Electro Chemical)

- ✓ The mechanical displacement to electrical energy transduction process takes place in the inner hair cells
- ✓ Bending of the inner hair cell cilia due to basilar membrane displacement produces a change in the overall resistance (reduces it) of the inner hair cell, thus modulating current flow through the hair cell.
- ✓ The modulation being directly proportional to the degree of bending of the cilia and the bending of the cilia is one direction only; in effect a half wave rectification of the basilar membrane displacement takes place.
- ✓ Bending of the cilia releases neurotransmitter which passes into synapses of one or more nerve cells which fire to indicate vibration
- ✓ The amount of firing is thus related to the amount of vibration
- ✓ Since the neurotransmitter is only released when the cilia are bent in one direction, firing tends to be in phase with basilar membrane movement




Inner Hair Cell model




Here bending the inner hair cell cilia is simulated by charging of the capacitor and returning to the initial position of the cilia is equivalent to discharging the capacitor.

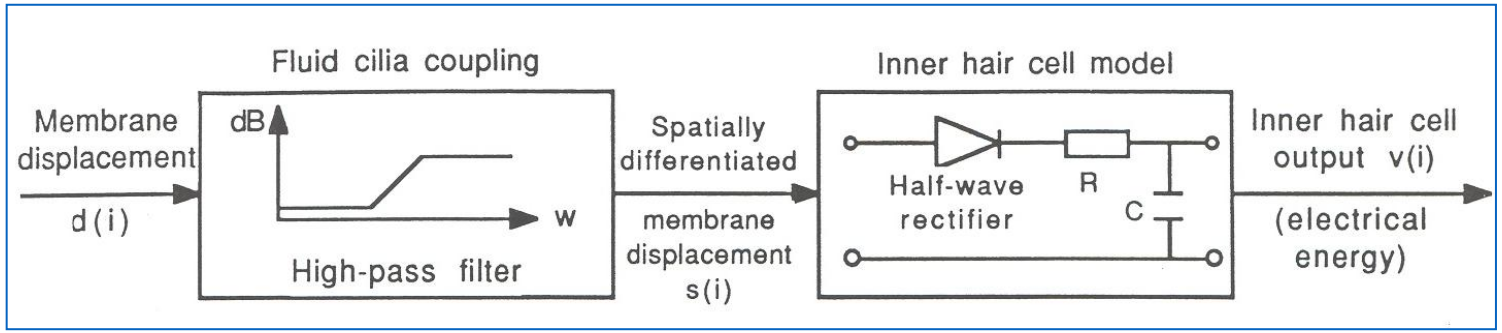
# Mechanical to Neural Transduction (Electro Chemical)

- ✓ The model inner hair cell is a capacitor model, in which the input voltage corresponds to the spatially differentiated membrane displacement output of the auditory model. (Second part of Figure below). Here bending the inner hair cell cilia is simulated by charging of the capacitor and returning to the initial position of the cilia is equivalent to discharging the capacitor.
- ✓ Spatially differentiation refers to taking the derivative with respect to position (along the basilar membrane) and a discrete model is given by:

$$s[i] = \frac{d[i+1] - d[i]}{\Delta x_i}$$

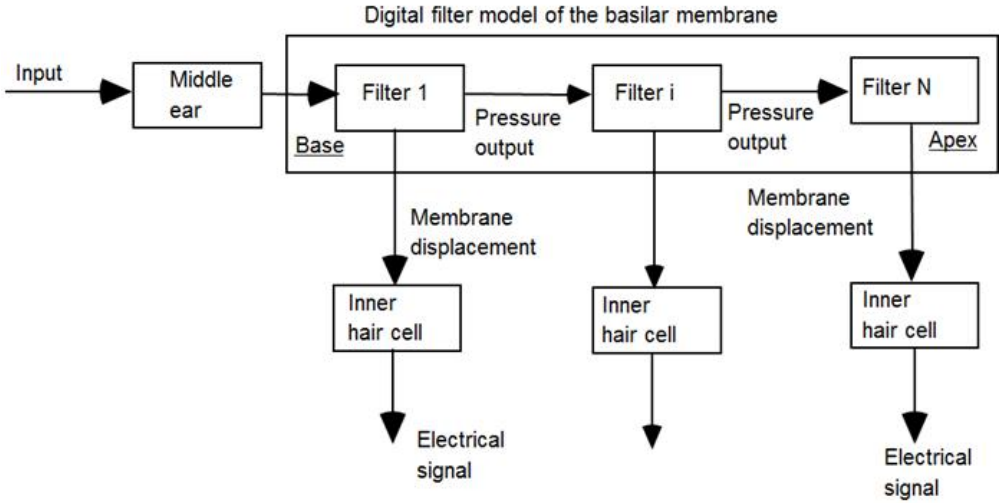
where, $d[i]$ is the displacement at the $i^{th}$ section along the membrane and $\Delta x_i$ is the width of the $i^{th}$ section

- ✓ Spatial differentiation of the membrane displacement represents coupling between the cilia of the inner hair cells , through the fluid in the subtectorial space ( high-pass filter effect, first part of Figure below)
- ✓ You will implement a digital model for neural transduction in addition to the transmission line auditory model

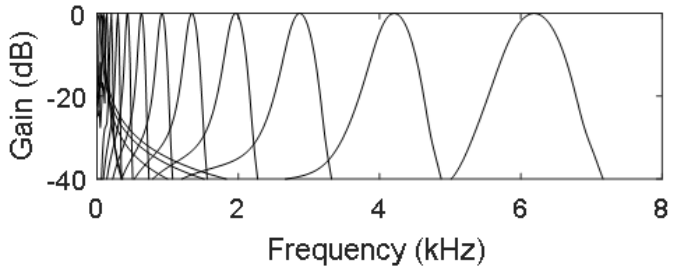


# Cochlear Modelling: Cascade and Parallel Models

## Transmission Line Model

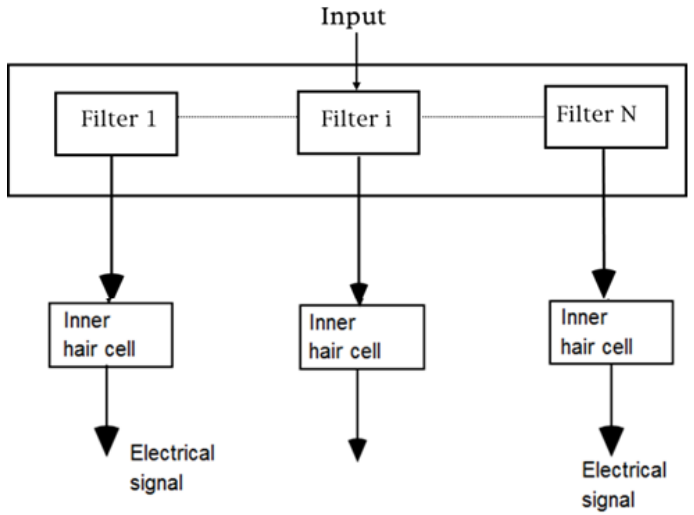



- ✓ The basic cochlear model shown above is a transmission line (cascade) model in which the basilar membrane is represented by 512 cascaded digital filter sections, each combining a resonator and a notch filter.
- ✓ Each digital filter section represents a segment of the basilar membrane tuned to a specific frequency. Together, the 512 sections model the entire basilar membrane.

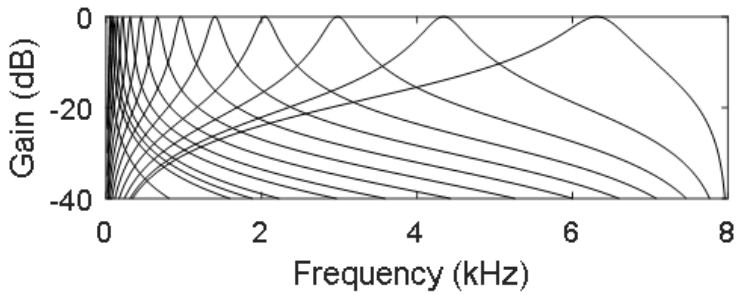



**Magnitude response of the cascaded filter bank model**

## Parallel Filter Bank Model

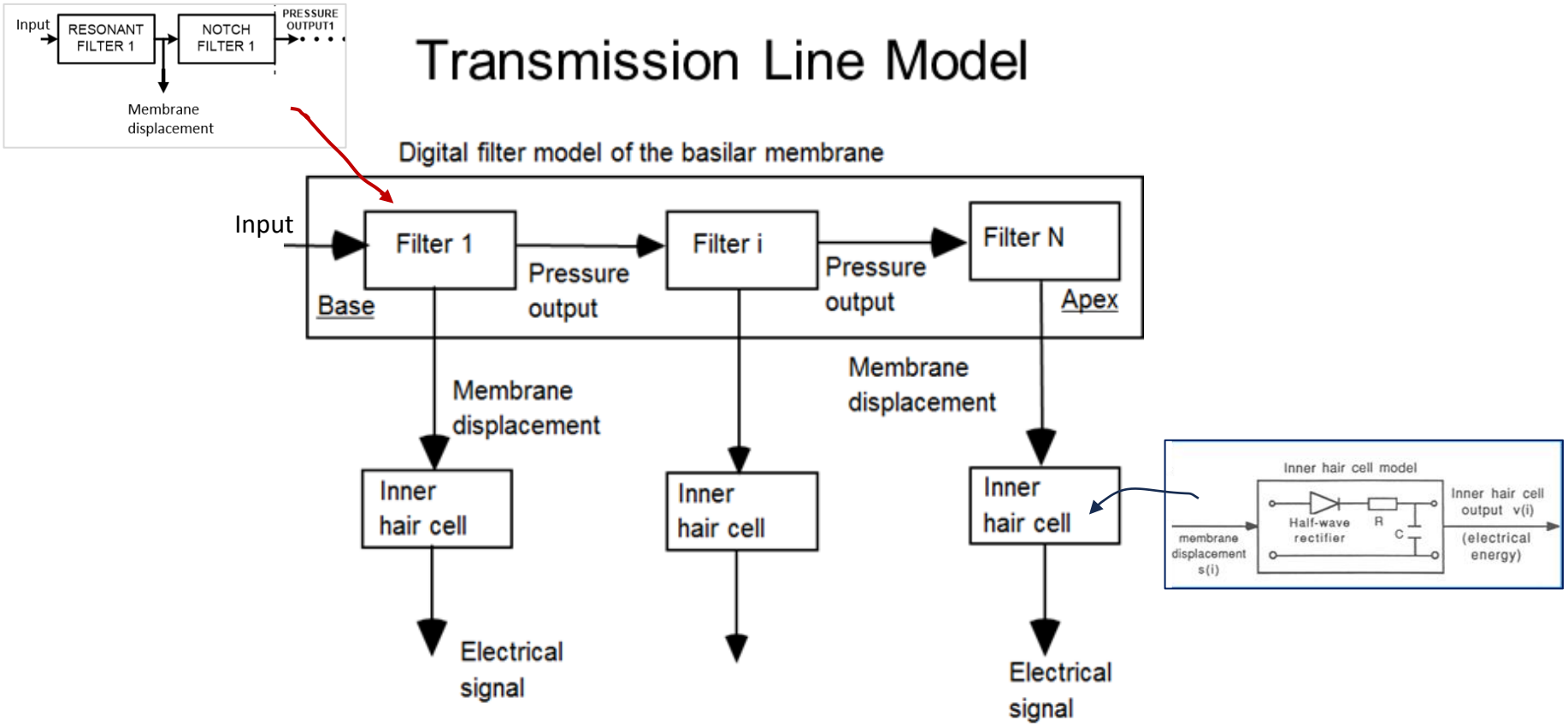



- ✓ In this model, the peripheral auditory system is represented by a bank of 512 bandpass filters (**auditory filters**) with overlapping passbands.
- ✓ Each filter corresponds to an equally spaced position along the basilar membrane and is tuned to the characteristic frequency at that position.




**Magnitude response of the parallel filter bank model**

# Cochlear Modelling: Transmission Line Model




In this mini-project, we will implement a transmission line (cascade) model of the cochlea using 512 cascaded digital filter sections. Each section combines a resonator and a notch filter and represents a segment of the basilar membrane tuned to a specific frequency.